SURESH KUMAR

FOUNDATION FOR INNOVATION AND TECHNOLOGY TRANSFER (FITT)-SCREENING

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Name
Suresh kumar

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Contact Number
+91 9877242938

Country, State and City
Greater Noida, Uttar Pradesh, India

Type of Applicant
Individual

Founder Type
Other

Whether applying through any FITT Program(s)
Yes

Please specify name of the Program
NIDHI-PRAYAS

Domain
Green Tech/Sustainability

Current Stage
Prototype

TRL
TRL 2

Clearly define the problem/gap you are addressing.
Waste-to-energy lacks decentralized, low-cost solutions. Current industrial reactors are expensive and emission-heavy. My project bridges this gap with a 6-layer, zero-electricity reactor that captures carbon emissions while producing fuel.

Why does this problem remain unsolved or poorly solved today?

Existing tech is centralized, high-cost, and complex, requiring huge capital. Small-scale solutions struggle with heat efficiency, hazardous emissions, and technical complexity, making them economically unviable for local waste-to-fuel conversion.

Existing solutions and their limitations.

Large-scale incinerators are costly, energy-intensive, and generate toxic flue gases. Small-scale manual systems fail to manage harmful emissions or sustain temperatures for efficient conversion, leading to low yield and environmental hazards.

Describe Your Technology

A modular 6-layer reactor using low-cost insulating materials. It utilizes a custom steam-scrubbing subsystem to capture gaseous emissions, converting them into industrial-grade sodium carbonate while producing synthetic fuel from biomass/waste.

What is the inventive step or novelty compared to existing solutions?

It integrates a high-efficiency steam-scrubbing system into a modular, multi-layer reactor. Unlike costly industrial plants, this design enables decentralized waste-to-fuel conversion with direct carbon capture at a fraction of the cost.

Who is the end user or customer?

Local waste management agencies, municipal bodies, and small-scale farms. They need a cost-effective, decentralized way to process biomass waste into fuel while actively neutralizing harmful greenhouse gas emissions on-site.

Current or expected competitors

Large-scale waste-to-energy plants and industrial incinerator manufacturers. Most current solutions are massive, high-capex systems. Smaller decentralized players lack the integrated emission scrubbing/carbon capture tech I provide.

IP status (Filed / Granted / In-process / Not filed)

Not filed

Describe your team’s roles, time commitment, and execution capability, including key advisors or collaborators

I am the Lead Applicant and Project Manager, responsible for the project's strategic and administrative oversight. My son is the Inventor; he leads the technical research, design, and prototyping. We dedicate 20+ hours weekly to this venture.

Are you looking for Incubation support from FITT?

Yes

Have you previously applied for FITT Incubation?

No

What support do you seek from FITT, and how will it help your startup? (e.g., incubation space, expertise, facilities, IP support, mentoring, etc.)

I seek access to FITT's advanced labs and testing facilities to scale our reactor prototype. Mentorship and IP support are crucial for navigating commercialization, while incubation space will provide the professional environment to drive growth.

Please describe the major technical, business, and commercialization milestones to be achieved over 2 years of incubation at FITT

Year 1: Scale the reactor to a pilot unit, validate thermal efficiency, and file IP.
Year 2: Build a functional demonstration unit, conduct rigorous performance testing, and partner with waste management firms to initiate commercial market trials.

Upload Pitch Deck (only in PDF format)

https://api-legacy.accubate.app/document/fitt/910a8a79-69ec-46b7-b60f-4e47b8d0a573

Declaration

I have read and agree to the above declaration.

Email
gyanjawala@gmail.com

Name
Suresh kumar

Gender
Male

Product Title or Brand Name
GBCM

Association with IIT Delhi
IIT Faculty

Social Category
SC

Does Company have Woman Founder
No

Founder(s) Background

Founder(s) Name	Highest Qualification	College/University	Work Experience
Rohit Kumar	12th pass	No college	Years

Registered Company Address
None

DPIIT Number

Year of Incorporation of Company
2027

Startup Stage
Proof of Concept

Paid up of the Company (INR)
0

Total Number of Shares of the Company

0

Shareholding Pattern

Shareholder Name	Share Percentage	Number of Shares
None	0	0

TRL

TRL 2

Company Website

Company LinkedIn Page

Founder LinkedIn Profile

Your ask, i.e. your expectation from FITT Incubation (what kind of support you require?)

Research & Development,Funding,Industry Connects